Russia: National Innovation Strategy Brief

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Country assessment: Russia's national innovation system

Russia is a scientifically capable country whose innovation system underperforms its human capital. In WIPO's Global Innovation Index (GII) 2025, Russia ranks 60th, far behind the United States (3rd), China (10th), and France (13th), and it has fallen from 45th in 2021 [2]. Its profile is lopsided: human capital and research is its best pillar (28th), while institutions rank 131st, by far the weakest of the four game countries. Russia also ranks higher on innovation outputs (55th) than on inputs (73rd), meaning it gets comparatively more out of what it puts in [2].

Indicator	Russia	United States	China	France
GII 2025 overall rank [2]	60	3	10	13
Human capital and research, rank [2]	28	13	20	15
Knowledge and technology outputs, rank [2]	62	3	1	15
Institutions, rank [2]	131	16	44	33
R&D spending, % of GDP [3]	1.10 (2020)	3.59 (2022)	2.56 (2022)	2.22 (2021)
Education spending, % of GDP [1]	4.2 (2023)	5.4 (2021)	4.0 (2023)	5.3 (2022)
Real GDP, PPP, $ trillion, 2024 [1]	6.09	25.68	33.60	3.73

Education. Russia's educational base is broad: literacy is 99.9% and school life expectancy is 15 years [1]. Education spending (4.2% of GDP) is close to China's but below the United States and France. Human capital is the one dimension where Russia is within range of its rivals, so it is the asset to protect and deploy. Spending shares measure inputs, not quality, and the years differ across countries.

Technology. Russia spends about 1.1% of GDP on R&D, roughly a third of the U.S. rate and less than half of China's or France's [3]. Its knowledge and technology outputs rank 62nd, against China's 1st [2]. Capability is concentrated in legacy sectors such as space, where Russia runs one of the world's largest programs [1]. EU export restrictions on dual-use and advanced technologies further limit access to foreign components [4]. The game reflects this legacy: Russia starts with +7 tacit knowledge (TK) versus France's +4 and is credited with "late-game disruptive potential," but it has the lowest wealth (W) [8].

Intelligence and immigration. Russia has a sophisticated state intelligence apparatus; the UK government attributes extensive cyber operations to military intelligence (GRU) units such as Unit 26165 [5]. Its main talent problem is skilled emigration: about 100,000 IT specialists, roughly 10% of the sector, left in 2022 by the government's own count [6]. Aggregate net migration is still slightly positive (+0.76 per 1,000) [1], so population totals hide this outflow. In the game Russia starts closed and "vulnerable to talent outflow," while the United States and France start open and are "more vulnerable to espionage" [8].

Implication. Russia cannot win a spending race. It should turn existing knowledge into output efficiently and build toward a late breakout. Because the final score rewards distance traveled as well as technology points and unspent wealth [8], the lowest-starting country has the most ground to gain.

Education recommendation (Tiger)

Russia's real-world strength is breadth of education: near-universal literacy, 15 years of expected schooling, and a human-capital-and-research ranking (28th) far above its overall innovation ranking (60th) [1, 2]. The weakness is conversion, not schooling. Education spending trails the United States and France, and weak institutions limit how well trained people are used. In the game, education matters through one channel: secondary (SE) and tertiary (TE) education raise human capital (K), which improves the success rate of every basic and applied research roll [8].

I recommend putting about 15% of the Round 1 budget into education, front-loaded in the first two rounds and tapered afterward. Because K improves every later roll, a unit bought in Round 1 pays off for the rest of the game, while a unit bought in the last round or two may never pay back; with 3-7 undeclared rounds, early is the only safe time to invest. While immigration stays closed, TE's extra benefit (raising K among international visitors) does not apply, so SE and TE compete only on K gained per point. We should buy whichever is cheaper per unit of K on our budget counter rather than split the money mechanically. Unlike France, Russia has no cheap-SE advantage, and unlike China and the United States, it has no strong education starting point [8].

This strategy is better than either extreme. Neglecting education wastes Russia's strongest real-world asset and lowers our odds on the medium- and high-risk research that our late-game plan depends on. Over-investing would starve applied research, our main source of points, in a country with the smallest budget. Real-world education policy pays off over decades; because the game may last only a few rounds, our game strategy instead treats education as a short, early investment in research success. If K gains turn out to be small relative to their price, I will shift the education share into research or savings after Round 1.

Technology strategy recommendation (Steve)

Russia's real-world technology base is narrow but deep. R&D intensity is about 1.1% of GDP, far below every rival [3], and EU export controls cut off many advanced inputs [4]. Moscow's response has been to concentrate state resources in legacy strengths such as space, nuclear, and defense, and to pursue import substitution. GII 2025 still shows Russia producing more output than its inputs predict [2]. The game has no sanctions, so we do not need to copy import substitution, but the main lesson carries over: concentrate on what Russia already knows how to do.

I recommend a "build, then break out" strategy. In Round 1, put 45% of the budget into low-risk applied research (L), the only category initially available, because our +7 TK raises success odds and earns early points and distance traveled. Put 20% into basic research (BR), which sets the minimum requirement for medium (M) and high (H) research. Foreign BR is public and counts toward ours at half strength, and the United States, with the highest budget, is likely to invest heavily [8]. We should therefore track rivals' public BR each round and buy only enough of our own to cross the next threshold. Our own BR also earns tacit knowledge, which only the investing country receives, and that widens our TK lead.

Once M unlocks, I will move spending from L to M whenever M offers more expected points per spending point. By Round 3, earlier if the BR threshold is met, we should open a bounded H position to exploit Russia's late-game disruptive potential. A game of 3-7 undeclared rounds means an H program started too late may never pay off. I will keep a floor of L spending so that one failed H roll cannot leave us with nothing. Gilli and Gilli argue that complex technologies depend on tacit and organizational knowledge that cannot simply be copied [7]; Russia's TK advantage is exactly that kind of knowledge, and it is the basis for pursuing our own breakthrough rather than imitation.

Intelligence and immigration recommendation (Gloria)

In the real world, Russia combines strong intelligence capabilities with a talent-retention problem. The UK government attributes sophisticated cyber operations to GRU units [5], while about 100,000 IT specialists left in 2022 [6]. Russia's weak institutions (131st) make it hard to attract or win back skilled workers [2]. Russia's real posture is high research security and restricted openness, and the game starts it in the same place: closed immigration with a slight defensive bonus against espionage [8].

I recommend keeping immigration closed. Under the rules, inflow from openness depends on wealth and relative university quality [8]. Russia has the lowest W and no TE advantage, so opening would likely attract little talent while giving up our espionage defense. The United States and France begin open and are the two countries the rules list as "more vulnerable to espionage," and Russia gains by not becoming a third soft target. The rules do not say whether closure stops outflow, so I will ask the instructor and reconsider openness only if our TE rises well above the rivals'.

For espionage, I recommend no spending in Round 1 and a growing program afterward. A critically high roll earns all of the target's points in a category since the start of the game [8], so the prize is near zero in Round 1 and grows every round. From Round 2 or 3, we should spy on the United States, which has the largest budget and the most applied research to steal, and on France, the other vulnerable country, using our two allowed targets. Ordinary successes reveal a target's applied-research spending history, which tells us which category is worth stealing and whether rivals are moving into M or H. I propose capping espionage at 10-15% of a round's budget so that it supplements our own research rather than replacing it. Critical failures expose us to all players, so we should pause after one and reassess.

Executive recommendation (Steve)

Decision rule. Each czar proposes the allocation for their portfolio. All three members vote, and two votes adopt a decision. As executive, I break ties and decide if the time limit (10 minutes for the opening, 5 minutes per round) expires before agreement. Note-taking and statistics duties rotate among all three members, as the instructor asked.

Overall strategy and Round 1 allocation. Our strategy is to build early points and capabilities, then break out late. It targets all three parts of the final score: technology points from L research, unspent wealth from a standing reserve, and distance traveled, where Russia's low starting position is an advantage. Round 1 shares of new spending points are: 45% L applied research, 20% basic research, 15% education (SE or TE, whichever yields more K per point), 0% espionage, and 20% saved, which carries over with interest. Immigration stays closed. We will convert these shares into whole purchases using the budget counter and Russia's actual prices.

Adjustments and consensus. After Round 1, we will move L spending into M once it unlocks and pays better, open a bounded H position by Round 3, start espionage against the United States and France from Round 2 or 3, and taper education after Round 2. We will change course if a revealed rule, price, or rival move changes these returns, and we will not spend the reserve in the final round unless doing so clearly beats the score value of keeping it. **[Complete after the team meeting: state whether all three members agreed on each recommendation, any dissent, and how the voting rule resolved it.]**

References

[1] Central Intelligence Agency. *The World Factbook*, entries for Russia, United States, China, and France (archived data, reproduced at theworldfactbook.org after the CIA retired the publication in February 2026). Accessed November 2026.

[2] World Intellectual Property Organization. *Global Innovation Index 2025* country profiles: Russian Federation, United States, China, France.

[3] National Science Board. *Science and Engineering Indicators 2025* (NSB-2025-7), Table DISC-10: GERD, GDP, and GERD-to-GDP ratio by selected country. Russia's latest reported year is 2020.

[4] European Commission. *Export-related restrictions for dual-use goods and advanced technologies*, FAQ on Council Regulation (EU) No 833/2014, updated September 25, 2026.

[5] UK Government. *Profile: GRU cyber and hybrid threat operations*, updated July 13, 2026.

[6] Interfax. "About 100,000 IT specialists left Russia in 2022 - digital development minister," December 20, 2022.

[7] Gilli, Andrea, and Mauro Gilli. 2019. "Why China Has Not Caught Up Yet: Military-Technological Superiority and the Limits of Imitation, Reverse Engineering, and Cyber Espionage." *International Security* 43(3): 141-189. doi:10.1162/isec_a_00337.

[8] Canfil, Justin K. *SIMULATION RULES* and *SIMULATION PREP*, course materials, 2025.